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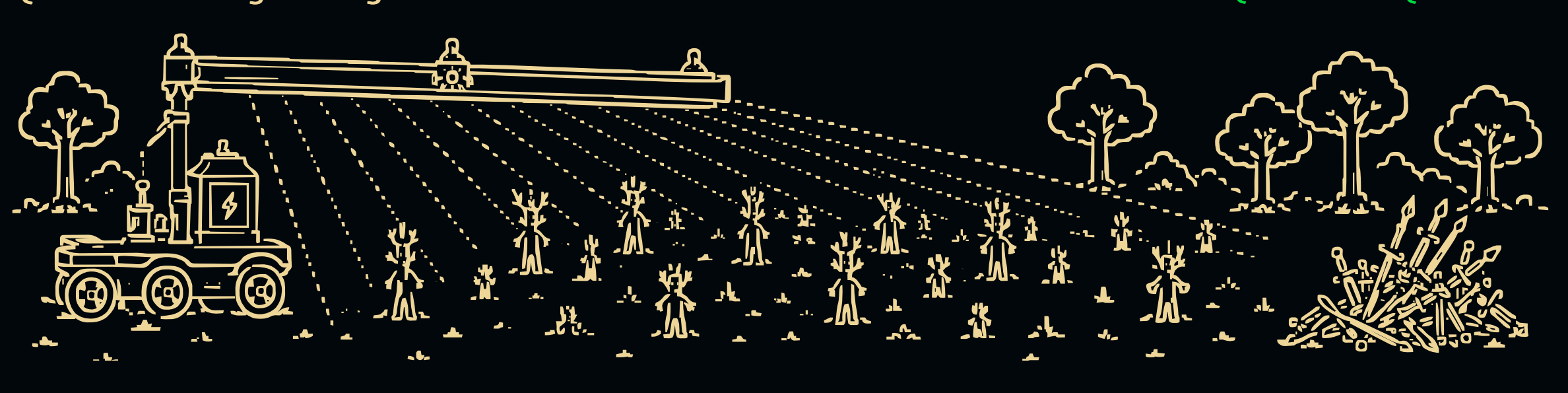
Lord of the Wings [2026]

Quest 8: The Entgineering

‹ Quest 7 Quest 9 ›



Part I

Story section

Sarumonk was finally defeated. With one last furious look, he disappeared into a demonic portal that closed behind him without a trace. At the same moment, the cage opened and the rest of the team was free.

We could not celebrate for long. The contaminated food was still spreading through the kingdom, so we needed help from someone who did not have to worry about eating it. That left surprisingly few options. I suggested asking the forest folk for help. Deep in the old woods lived the Ents, strange tree-like creatures with very little interest in bread, meat pies or turnip stew. For once, being made mostly of wood seemed like a considerable advantage.

Frodoff was also getting worse. His eyes often seemed distant, as if he were looking at something none of us could see, and at night he sometimes woke suddenly with a scream.

When we finally reached the Ents and explained what had happened, they agreed to help. Better still, they already had a plan. If the kingdom needed defenders who could not be poisoned by ordinary food, they would simply grow more Ent warriors.

The Ents had prepared special growing fields made of hexagonal plots. Each occupied plot contained one young Ent, whose height was written as a number from `1` to `9`. They gave me a map of one of the fields (your notes).

Growing an army required a continuous supply of energy, so the Ents first needed to know how much energy their field would consume. Their energy source worked much like sunlight: every Ent reached by it required **one unit of energy per round**.

For the first test round, the energy source stayed in one position and sent energy from the upper left towards the lower right. Because the energy travelled across the field at an angle, the Ents could cast shadows over one another. **The shadow line descended across the field, losing one level of height with every hex it crossed.** For example, an Ent of height `6` shaded the following hexes up to heights `5`, `4`, `3`, `2` and `1`, before its shadow disappeared completely.

If an Ent was no taller than the shadow reaching its hex, no energy reached it. A taller Ent rose above that shadow and still received energy. Every Ent cast its own shadow according to its height, even if that Ent was itself already in shadow.

I needed to determine **how many units of energy were required for this test round**.

Example based on the following notes:

```
....1.9.6.1.7....
...9.2.3.3.8.1...
..7.8.3.5.1.4.3..
.4.6.7.4.1.1.8.9.
4.5.3.6.5.8.9.4.9
.9.5.1.5.6.2.8.9.
..7.7.6.4.6.2.4..
...9.3.8.3.8.4...
....1.6.6.2.8....
```

Each digit represents the height of one Ent on the hexagonal field. The dots are only used to preserve the layout of the hex grid. During this round, the energy travels diagonally across the field from the upper left to the lower right.

The first line of Ents reached by the energy is:

```
Direction: ↘

4....
.9...
..7..
...9.
....1
```

The first Ent, with height `4`, receives energy directly. Its shadow reaches the following hexes at heights `3`, `2` and `1`.

The next Ent has height `9`, so it rises above the shadow of height `3` and also receives energy. Its much larger shadow continues with heights `8`, `7`, `6`, and so on.

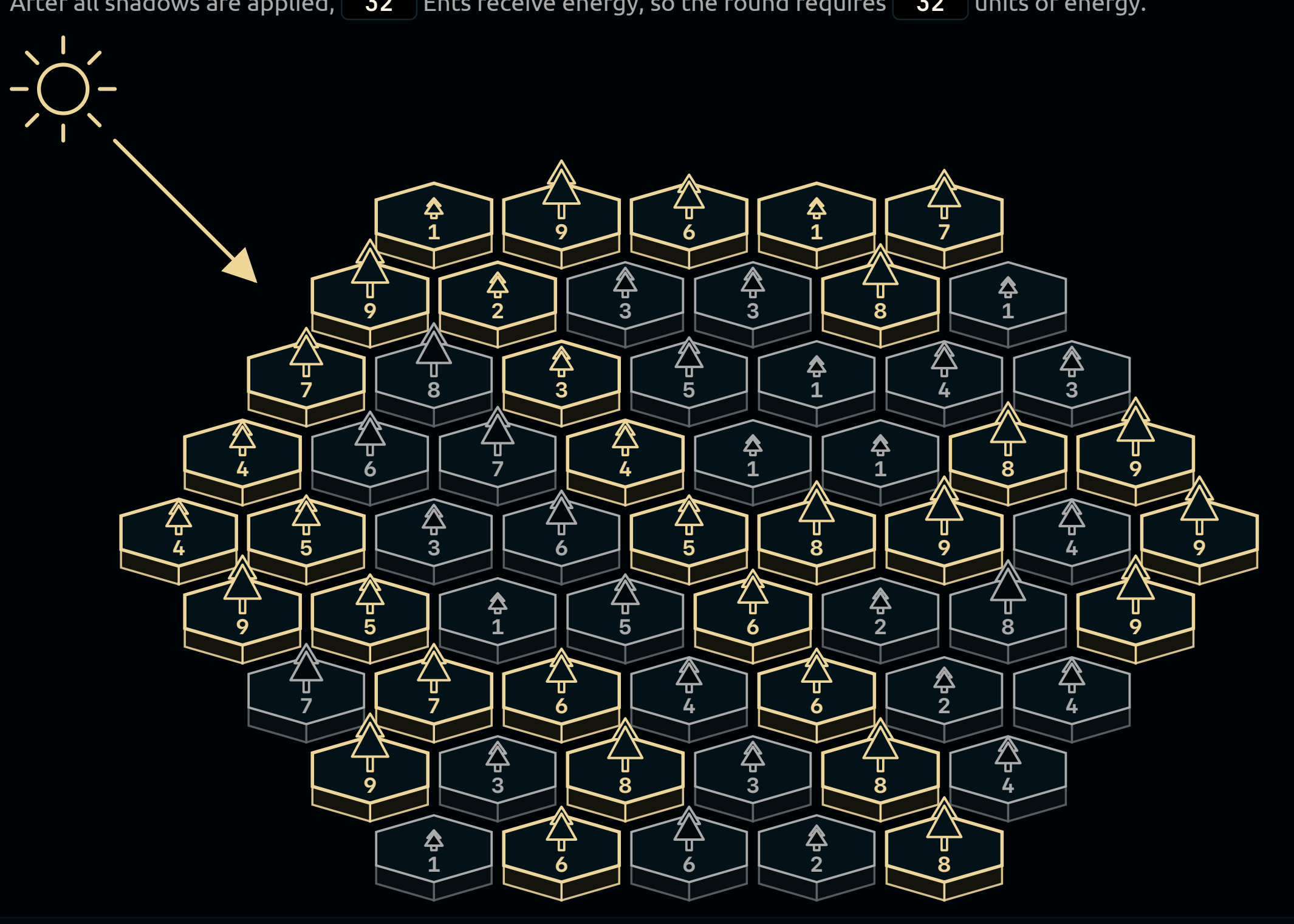
The Ent of height `7` is therefore hidden by a shadow of height `8`. The following Ent of height `9` rises above the shadow reaching it and receives energy, while the final Ent of height `1` is hidden by its shadow.

With every shaded Ent marked as `.`, the complete map is:

```
Direction: ↘

....1.9.6.1.7....
...9.2.....8.....
..7...3..........
.4.....4.....8.9.
4.5.....5.8.9...9
.9.5.....6.....9.
....7.6...6......
...9...8...8.....
......6.....8....
```

After all shadows are applied, `32` Ents receive energy, so the round requires `32` units of energy.



How many units of energy were required for this round?

Your notes for this part: Copy Open Download

Part I solved with answer: **171**
Check your progress

Part II

Everything seemed simple enough until the Ents showed us one of their real growing fields. The energy source was not fixed in one place. Instead, it travelled around the entire field.

A complete circuit consisted of `6` rounds. In each round, the source occupied the next position and sent energy along one of **the six straight directions of the hexagonal grid**: `↘`, `↙`, `←`, `↖`, `↗` and `→`.

The rules for shadows remained exactly the same. During this test, the same field was used for all six rounds. As before, every Ent reached by the energy required **one unit of energy in that round**.

Gimlock immediately declared that such an unnecessarily complicated mechanism had obviously been designed by Elves. Legolink denied this so firmly that I decided not to ask any further questions.

The Ents needed to know how much energy one complete circuit required. Using the growing field (your notes), I needed to **count the Ents receiving energy in each of the six rounds and add those six values together**.

Example based on the following notes:

```
....1.9.6.1.7....
...9.2.3.3.8.1...
..7.8.3.5.1.4.3..
.4.6.7.4.1.1.8.9.
4.5.3.6.5.8.9.4.9
.9.5.1.5.6.2.8.9.
..7.7.6.4.6.2.4..
...9.3.8.3.8.4...
....1.6.6.2.8....
```

The same field is used for all six rounds of one complete circuit. The number of Ents receiving energy, and therefore the energy required in each round, is:

```
Round: 1
Direction: ↘
Energy: 32

....1.9.6.1.7....
...9.2.....8.....
..7...3..........
.4.....4.....8.9.
4.5.....5.8.9...9
.9.5.....6.....9.
....7.6...6......
...9...8...8.....
......6.....8....
```

```
Round: 2
Direction: ↙
Energy: 32

....1.9.6.1.7....
...9.....3.8.1...
....8...5...4.3..
.....7.......8.9.
......6...8.9...9
.9.....5.....8.9.
..7.7.6..........
...9...8...8.....
........6...8....
```

```
Round: 3
Direction: ←
Energy: 29

......9.6...7....
...9.......8.1...
....8...5...4.3..
.....7.........9.
..5.........9...9
.9.5...........9.
..7.7.6...6...4..
...9...8...8.4...
......6.....8....
```

```
Round: 4
Direction: ↖
Energy: 32

......9.6...7....
...9.......8.....
..7.8............
...6.7.......8.9.
......6...8.9...9
.9.....5.....8.9.
....7...4.....4..
...9...8.3.8.4...
....1.6.6.2.8....
```

```
Round: 5
Direction: ↗
Energy: 30

......9.6........
...9.......8.....
..7.8...5........
.4...7.........9.
4.......5.8.9...9
.9...........8.9.
..7...6...6......
...9.3.8...8.....
....1.6.6.2.8....
```

```
Round: 6
Direction: →
Energy: 32

....1.9.....7....
...9.......8.....
..7.8............
.4.6.7.......8.9.
4.5...6...8.9...9
.9.......6...8.9.
..7.7.....6......
...9...8...8.....
....1.6.6...8....
```

The energy required for the complete six-round circuit is therefore:

32 + 32 + 29 + 32 + 30 + 32 = **187**

How many units of energy were required for one complete six-round circuit of the energy source?

Your notes for this part: Copy Open Download

Part II solved with answer: **1001**
Check your progress

Part III

The tests were complete, but the Ents needed one final calculation before starting continuous production. They had to know how much energy would be required to grow enough warriors.

In the actual process, the field changed after every round. **Each Ent receiving energy grew by one level.** When an Ent of height `9` received energy, it became a fully grown warrior. It left the field and joined the army, leaving a seed behind in its place.

The seed remained underground for the next `3` rounds, required no energy and cast no shadow during that time. After remaining underground for three complete rounds, it sprouted as a new Ent of height `1` at the start of the following round. From then on, it followed the normal growing rules again.

The energy source continued its six-round circuit exactly as before. In every round, it moved to the next position, and every Ent reached by the energy required one unit of energy. All the rules for shadows remained unchanged, and all shadows were determined before any Ents grew during that round.

The Ents estimated that the mission needed an army of at least `100000` fully grown warriors. Production therefore continued until at least that many Ents had left the field and joined the army.

Using the growing field (your notes), I needed to simulate the complete production process and **add the full energy required in every round, including the round in which the army first reached at least `100000` warriors.**

Example based on the following notes:

```
....1.9.6.1.7....
...9.2.3.3.8.1...
..7.8.3.5.1.4.3..
.4.6.7.4.1.1.8.9.
4.5.3.6.5.8.9.4.9
.9.5.1.5.6.2.8.9.
..7.7.6.4.6.2.4..
...9.3.8.3.8.4...
....1.6.6.2.8....
```

In every round, the energy source moves to the next of its six positions. Every Ent receiving energy uses one unit of energy and grows by one level. An Ent of height `9` that receives energy leaves the field as a new warrior, and its seed remains underground for three rounds before sprouting again at height `1`.

The first ten rounds are shown below. Each map shows the field at the start of the round. Ents receiving energy are marked in **bold**. The symbol `X` represents a seed that is still underground and does not require energy.

```
Round: 1
Direction: ↘
Energy: 32
New warriors: 8

....1.9.6.1.7....
...9.2.3.3.8.1...
..7.8.3.5.1.4.3..
.4.6.7.4.1.1.8.9.
4.5.3.6.5.8.9.4.9
.9.5.1.5.6.2.8.9.
..7.7.6.4.6.2.4..
...9.3.8.3.8.4...
....1.6.6.2.8....
```

```
Round: 2
Direction: ↙
Energy: 31
New warriors: 6

....2.X.7.2.8....
...X.3.3.3.9.1...
..8.8.4.5.1.4.3..
.5.6.7.5.1.1.9.X.
5.6.3.6.6.9.X.4.X
.X.6.1.5.7.2.8.X.
..7.8.7.4.7.2.4..
...X.3.9.3.9.4...
....1.7.6.2.9....
```

```
Round: 3
Direction: ←
Energy: 29
New warriors: 5

....3.X.8.3.9....
...X.4.3.4.X.2...
..9.9.4.6.1.5.4..
.5.6.8.6.1.1.X.X.
5.6.3.7.6.X.X.5.X
.X.7.1.5.7.2.9.X.
..8.9.8.4.8.2.5..
...X.3.X.3.X.5...
....1.7.7.2.X....
```

```
Round: 4
Direction: ↖
Energy: 22
New warriors: 4

....3.X.9.3.X....
...X.5.3.5.X.3...
..X.X.4.7.1.6.5..
.5.6.9.7.2.2.X.X.
5.7.3.8.7.X.X.6.X
.X.8.1.5.7.2.X.X.
..8.X.9.4.9.2.6..
...X.4.X.3.X.6...
....1.8.8.3.X....
```

```
Round: 5
Direction: ↗
Energy: 25
New warriors: 5

....3.1.X.3.X....
...1.5.3.6.X.3...
..X.X.4.8.1.7.6..
.5.6.X.8.2.2.X.1.
5.7.3.9.7.X.1.7.1
.1.9.1.6.7.2.X.1.
..9.X.X.5.X.2.7..
...1.4.X.4.X.7...
....2.9.9.4.X....
```

```
Round: 6
Direction: →
Energy: 27
New warriors: 1

....4.1.X.3.X....
...1.6.3.6.1.3...
..X.X.4.9.1.8.7..
.6.6.X.8.2.2.1.1.
6.8.3.X.8.1.1.8.1
.2.X.2.7.8.2.X.1.
..X.X.X.5.X.2.8..
...2.5.1.4.1.8...
....3.X.X.5.1....
```

```
Round: 7
Direction: ↘
Energy: 28
New warriors: 8

....5.1.X.4.1....
...2.7.3.7.1.3...
..1.1.5.X.1.9.7..
.7.7.X.9.2.2.1.1.
7.9.3.X.9.1.1.9.1
.3.X.3.8.9.2.1.1.
..X.1.X.6.X.2.9..
...3.6.1.5.1.9...
....4.X.X.6.1....
```

```
Round: 8
Direction: ↙
Energy: 27
New warriors: 1

....6.2.1.5.2....
...3.8.4.8.1.4...
..2.1.5.X.1.X.8..
.8.8.1.X.3.2.1.1.
8.X.3.X.X.1.1.X.1
.3.X.3.9.X.3.1.1.
..X.1.1.6.1.3.X..
...3.6.1.5.1.X...
....5.X.X.7.1....
```

```
Round: 9
Direction: ←
Energy: 31
New warriors: 6

....7.3.2.6.3....
...3.9.5.9.1.5...
..2.1.6.X.2.X.9..
.9.9.1.X.4.2.1.2.
9.X.3.1.X.1.1.X.2
.3.1.3.X.X.3.2.2.
..1.1.1.7.1.4.X..
...4.6.1.6.1.X...
....5.1.1.8.2....
```

```
Round: 10
Direction: ↖
Energy: 26
New warriors: 1

....8.3.2.7.4....
...3.X.5.X.1.6...
..2.1.7.1.2.X.X..
.X.X.1.X.5.3.1.3.
X.X.4.2.X.2.2.X.3
.4.1.4.X.X.4.3.3.
..1.1.1.8.1.5.X..
...4.7.1.7.2.X...
....5.1.1.9.3....
```

The army reaches `100000` Ent warriors after `32008` rounds. Additional simulation details are available **here**.

Across all these rounds, the field requires a total of `899889` units of energy.

How many units of energy were required to grow an army of at least `100000` Ent warriors?

Your notes for this part: Copy Open Download

Part III solved with answer: **899356**
Check your progress

Puzzle solved! Don't stop now!

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